

CAMPAIGN PERFORMANCE PREDICTION AND BUDGET OPTIMIZATION SYSTEM

MINI PROJECT REPORT

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CHENNAI - 600 117, INDIA
2026

ISBN No: 978-93-344-7370-4

BONAFIDE CERTIFICATE

Certified that this mini project report entitled "CAMPAIGN PERFORMANCE PREDICTION AND BUDGET OPTIMIZATION SYSTEM" is a Bonafide work carried out by **Harishwaran K, Mageshwaran H, Keerthivarman M** of the Department of Artificial Intelligence and Machine Learning, VELS Institute of Science, Technology & Advanced Studies (VISTAS), Chennai, in partial fulfilment of the academic requirements.

This report is submitted as part of the academic mini project in the Department of Artificial Intelligence and Machine Learning.

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Place: Chennai

Date: _____

ACKNOWLEDGEMENT

We express our sincere gratitude to VELS Institute of Science, Technology & Advanced Studies (VISTAS), Chennai, for providing the academic environment for this mini project.

We thank our project guide, **Mrs. V. P. Kanchana**, for guidance and support. We also thank the faculty members of the Department of Artificial Intelligence and Machine Learning.

We thank our teammates and everyone who contributed feedback, testing assistance, and encouragement during the development of the virtual try-on prototype.

We acknowledge the researchers whose work on contextual bandits and advertising budget optimization informed our literature review. We also acknowledge the supplied project materials and deployed applications used to describe the proposed system and its visible interface.

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ABSTRACT

Digital advertising plays a significant role in modern marketing, enabling businesses to reach target audiences through multiple advertising channels. However, campaign performance varies depending on campaign characteristics, channel selection, and budget allocation, making it challenging to utilize advertising resources effectively. This paper proposes a system that integrates machine-learning-based campaign performance prediction with constrained advertising budget optimization to support data-driven marketing decisions. Using historical campaign data, the system applies supervised learning techniques to analyze campaign characteristics and predict expected performance across different campaigns and advertising channels. The system employs a Gradient Boosting Regressor to predict campaign revenue based on relevant campaign characteristics. Based on the predicted outcomes, an optimization layer determines a suitable allocation of the available marketing budget while satisfying predefined budget constraints. The system also incorporates a what-if analysis feature that enables users to modify budget allocations or campaign parameters and examine their potential impact on predicted performance and resource distribution. A web-based dashboard presents prediction results, channel recommendations, optimized budget allocations, and simulation outcomes in an accessible format. By integrating predictive analytics, channel recommendation, constrained optimization, and scenario analysis, the proposed system aims to improve advertising resource utilization, minimize inefficient spending, and support informed digital advertising campaign planning.

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PROJECT GUIDE

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ABBREVIATIONS

API - Application Programming Interface; A/B - two-variant comparison; CPA - Cost Per Acquisition; CPC - Cost Per Click; CTR - Click-Through Rate; GB - Gradient Boosting; KPI - Key Performance Indicator; MAE - Mean Absolute Error; ML - Machine Learning; RMSE - Root Mean Squared Error; R-squared (R2) - coefficient of determination; SLSQP - Sequential Least Squares Programming; UI - User Interface.

CHAPTER 1 - INTRODUCTION

1.1 Project Overview

Digital advertising gives organizations access to search, social, display, and other media channels. A marketing team must decide how to divide a limited budget across campaigns whose returns depend on audience, creative, timing, channel, and spend. A static allocation or a rule based only on past totals can miss diminishing returns: spending more on one channel may produce progressively smaller gains while another channel still has room to grow.

This project proposes a decision-support system that joins two tasks. A supervised machine-learning model estimates campaign revenue from historical campaign characteristics. A constrained optimizer then uses the predicted response to recommend an allocation that respects a total budget and per-channel bounds. A web dashboard makes the inputs, recommendation, and scenario workflow visible. The revised abstract identifies a Gradient Boosting Regressor for revenue prediction and proposes what-if analysis as part of the system [4].

1.2 Problem Statement

Campaign managers need to compare alternative allocations before money is committed. Manual spreadsheets can calculate totals but do not learn a prediction function from historical data or automatically solve a constrained allocation problem. The project addresses the following question: given campaign context, historical performance information, and a fixed budget, how can the system estimate expected revenue and find a feasible allocation across supported advertising platforms?

1.3 Objectives

- Prepare historical campaign attributes for supervised prediction, with a clearly defined target and a leakage-safe feature set.
- Estimate expected campaign revenue using the Gradient Boosting Regressor specified in the final abstract.
- Use predicted response curves in a constrained budget-allocation step.
- Expose campaign setup, channel comparison, and what-if inputs in a web dashboard.
- Report actual model and optimizer results only after the backend and evaluation data are available.

1.4 Scope and Users

The intended users are marketing analysts and campaign planners who need a transparent recommendation for a proposed campaign budget. The deployed interface presents a campaign setup flow, a budget optimizer, analytics, an A/B comparison view, audience insights, a what-if simulator, a report view, and an ad creative workspace [7]. The core research scope is prediction plus allocation. The interface also contains demo areas; those are described as demonstrations rather than treated as evidence of live customer-level analytics.

1.5 Applications and Expected Benefits

- Compare predicted campaign responses across advertising channels before a budget is committed.
- Check that platform allocations sum to the chosen budget and remain within stated minimum and maximum values.
- Explore alternative budget envelopes with a what-if workflow.
- Present campaign context and model output in a single reviewable dashboard.

The expected benefit is better structured decision-making. The project does not guarantee that an allocation will improve revenue in production; that conclusion would require measured, out-of-sample evaluation and, ultimately, controlled campaign evidence.

1.6 Challenges

Challenges include limited or unrepresentative historical data, differences between training and deployment markets, delayed conversions, attribution error, changing auction prices, and missing campaign context. Feature timing is critical: information that becomes known only after a campaign runs cannot be used as an input for a genuine pre-campaign prediction. A mathematically feasible allocation can still be poor if the fitted revenue curves are inaccurate or if an objective omits risk and business rules.

1.7 Report Organization

Chapter 2 reviews advertising allocation research and the project's research gap. Chapter 3 describes requirements, architecture, prediction, and optimization. Chapter 4 documents the deployed interface and verified runtime observations. Chapter 5 summarizes the work and recommends further validation.

CHAPTER 2 - LITERATURE REVIEW

2.1 Advertising Allocation as a Learning Problem

Advertising returns are uncertain and may change with audience, season, creative, channel, and spend. Research approaches range from offline prediction and mathematical optimization to sequential learning methods that observe outcomes and update later decisions. The literature supplied for this project emphasizes a key distinction: predicting performance is not the same operation as allocating a budget. A useful decision-support pipeline must connect its estimate to an explicit constraint-aware decision rule [5, 6].

2.2 Selected Base Paper

Han and Gabor's AdKDD 2020 paper studies budget selection when campaign payouts are uncertain. Their contextual-bandit system predicts payouts from context and historical performance, extrapolates payout distributions to budgets not directly observed, and applies Thompson Sampling to balance exploration and exploitation [1]. The authors report a 22 +/- 10 percent improvement in mean Cost Per Acquisition over a previous allocation strategy. That reported result belongs to their production system at Lyft; it is not a result of this student project [1].

The paper is relevant because it links prediction to a budget decision. Its production-scale contextual-bandit loop is outside the current project's stated scope. The student prototype instead describes supervised revenue prediction followed by constrained optimization. It does not claim to reproduce online bandit learning, sequential experimentation, or automatic live bidding.

2.3 Other Related Work

Nuara, Trovo, Gatti, and Restelli formulate joint bid and daily-budget decisions for pay-per-click advertising as a combinatorial semi-bandit problem. Their work considers online decision-making across campaigns, with the bid and daily budget selected together [2]. Gigli and Stella study parametric bandits for performance marketing, including multi-ad-group campaigns and bootstrapped Thompson sampling [3]. These works demonstrate the breadth of the research area; they also show that production advertising decisions may include bidding, delayed outcomes, exploration, and online updates beyond a static allocation prototype.

2.4 Comparison and Research Gap

Aspect	Research literature	This project
Primary decision	Sequential budget and/or bid selection under uncertain returns	Predict campaign revenue, then optimize a fixed budget
Learning setup	Contextual or combinatorial bandit methods	Supervised Gradient Boosting model in the final abstract
Decision step	Explore/exploit or joint bid-budget optimization	SLSQP allocation with explicit total and platform bounds in the UI
Evidence	Empirical or production results reported by each paper	No verified project metrics available in the supplied materials
User interface	Not the central contribution of the cited methods	Dashboard, what-if, analytics, and demo pages

The project does not claim that advertising optimization is novel. Its practical gap is integration: present a student-accessible workflow that connects a revenue predictor to constrained allocation, exposes inputs and assumptions, and provides a what-if interface. The project reference document explicitly separates the research foundation from the proposed implementation [5].

2.5 Literature Review Summary

The selected base paper motivates the use of predictions in budget decisions. More recent work illustrates sequential and multi-channel alternatives [2, 3]. The project's contribution is a simplified, inspectable prototype. Its value must be judged on data quality, prediction error, feasibility, and whether an allocation improves outcomes when compared with a suitable baseline. Until those measurements are completed, claims remain design goals rather than demonstrated impact.

CHAPTER 3 - SYSTEM ANALYSIS AND DESIGN

3.1 Existing and Proposed System

A manual planning process often combines separate campaign reports, spreadsheet calculations, and marketer judgement. This approach can be hard to reproduce and may not satisfy platform-specific spending limits. The proposed system organizes a campaign definition, performance estimate, allocation, and scenario view into one application. The visible interface labels its backend as FastAPI ML and its optimization endpoint as POST /optimize [7].

3.2 Functional Requirements

ID	Requirement	Verification evidence
FR1	Enter campaign name, industry, brand, market, objective, duration, target, and budget	Campaign Setup page exposes these fields
FR2	Enforce total budget and platform bounds	UI states equality and per-platform limits
FR3	Generate a revenue prediction and channel allocation	Requires an available optimizer backend
FR4	Compare channel efficiency and allocations	Analytics page provides a comparison area
FR5	Run a hypothetical budget scenario	What-if page requires a baseline and backend
FR6	Present campaign summary for review or printing	Reports page presents an executive report view

3.3 System Architecture

PROJECT PIPELINE
1. Historical campaign data and campaign context
2. Data validation and feature preparation
3. Gradient Boosting revenue prediction
4. Platform response curves and constrained SLSQP optimization
5. Allocation, analytics, and what-if scenario output
6. Web dashboard and report presentation

Figure 3.1: Proposed prediction-to-allocation system architecture. Conceptual architecture based on the project proposal and visible application design.

The architecture separates prediction from allocation. Historical campaign records are cleaned and transformed into model features. The predictor estimates expected revenue for supported campaign context. An optimization layer evaluates channel-level response against the available budget and bounds. The user interface collects inputs and displays outputs. The supplied conference draft proposed this pipeline and cloud access; the deployed page currently exposes a FastAPI ML interface and labels Gradient Boosting and SciPy SLSQP components [6, 7].

The application's connection to the backend is an important boundary. The updated deployment completed the optimizer request during the latest inspection. This verifies the sample workflow, not model quality or production behavior.

3.4 Data Flow and Processing

- Input: campaign metadata and historical records. Proposed historical fields include spend, impressions, clicks, CTR, conversions, CPC, CPA, channel, and campaign type [6].
- Validation: check missing values, units, duplicate records, currency, date ranges, and category labels.
- Feature preparation: encode categories and derive only values that are available at prediction time.
- Prediction: apply the selected Gradient Boosting Regressor to estimate the target revenue.
- Optimization: evaluate platform response and solve for an allocation subject to total-budget and platform constraints.
- Presentation: display the allocation, predicted revenue, ratios, constraints, and scenario comparison.

Leakage control is essential. Spend, clicks, conversions, CPA, or CTR measured after a campaign may reveal the outcome being predicted. They should be included only when their timestamps establish that they are known at decision time, or be excluded from the pre-campaign prediction set. The final dataset, target column, and feature timing were not included with the supplied files, so this report does not claim a validated training design.

3.5 Prediction and Optimization Formulation

Let x denote campaign context and let f(x) estimate expected revenue. For m supported channels, let b_i be the budget assigned to channel i and R_i(b_i, x) its predicted response. The allocation can be written as maximizing the sum of expected channel responses subject to a fixed budget and lower and upper bounds:

Objective	**maximize $\Sigma_i R_i(b_i, x)$**
Budget equality	$\Sigma_i b_i = B$
Feasible bounds	$l_i \leq b_i \leq u_i$ for each supported channel

The deployed budget page describes a channel-specific logarithmic response curve and a Sequential Least Squares Programming (SLSQP) minimizer. A representative form displayed in the interface is $R_i(b_i) = \alpha_i \log(1 + \beta_i b_i) + \gamma_i$, which captures diminishing marginal returns as spend increases [7]. The response parameters are displayed by the UI; their derivation and calibration data were not included in the supplied materials. They are therefore implementation settings, not independently validated estimates.

3.6 Market Constraints Visible in the Interface

Market	Supported platforms	Minimum and maximum	Total budget
India	Google Ads, Meta Ads	Google: ₹5,000-₹25,000; Meta: ₹5,000-₹15,000	₹10,000-₹40,000

For the India scope the UI excludes TikTok Ads and says the platform allocations must sum exactly to the chosen total. These limits are configuration displayed by the application, not a general regulatory statement. The selected sample configuration is ₹25,000 across Google and Meta, with 30 days, a ₹75,000 target revenue goal, SaaS industry, Zoho brand, and Search Engine Marketing objective [7]. In the latest sample run, Google received ₹14,928 and Meta received ₹10,072; both amounts fall within the displayed platform bounds and sum to the configured ₹25,000. A feasible allocation requires the total budget to remain between the sum of the platform minimums and the sum of their maximums.

The total budget is a hard equality constraint: Google allocation plus Meta allocation must equal the campaign budget. Platform bounds prevent the optimizer from placing an infeasible amount on either channel. For India, Google must receive ₹5,000 to ₹25,000 and Meta ₹5,000 to ₹15,000, so a valid total must be between ₹10,000 and ₹40,000. The latest ₹25,000 sample allocation assigns ₹14,928 to Google and ₹10,072 to Meta. Both amounts fall within their respective ranges and sum exactly to the selected total. This 60/40 split depends on the campaign configuration, market, model response curves, and selected budget; it is not a fixed recommendation for every campaign. If a proposed total falls outside the combined feasible range, the system should report infeasibility rather than silently violate a platform bound. The revenue estimate is a model output, not observed sales, and requires validation against campaign data.

3.7 Main Modules and Requirements

Module	Responsibility	Current evidence
Campaign setup	Collect campaign metadata and budget	Rendered form and default values
Prediction and optimizer	Estimate revenue and allocate budget	UI describes Gradient Boosting and SLSQP; latest sample run returned ₹87,240
Analytics	Show allocation, revenue, ratios, and channel comparison	Page loaded; values awaited optimizer
What-if simulator	Compare baseline and alternative budget	Page requires baseline and live POST request
A/B testing and audience	Illustrate comparison and audience views	Explicit demo labels; audience data marked synthetic
Reports and ad studio	Present executive summary and ad preview	Pre-filled report and copy workspace visible

Non-functional requirements include clear constraint messages, consistent currency handling, reproducible model versions, responsive page layout, validation of user inputs, and transparent labeling of demo data. A production deployment should protect campaign data, limit access to uploaded datasets, log model versions and input dates, and make stale predictions visible. Since campaign performance and budget details may be commercially sensitive, cloud deployment should include access control and data-retention policies.

3.8 Software and Hardware Context

The deployed interface identifies a FastAPI ML service. The budget page names SciPy SLSQP. The available project material does not provide a complete dependency lock file, exact server version, training hardware, or dataset size. The website was inspected at the public Vercel deployment on 29 October 2026. The updated deployment completed an optimizer run during inspection on 30 October 2026, returning a sample predicted revenue of ₹87,240 for the ₹25,000 budget [7].

CHAPTER 4 - IMPLEMENTATION AND RESULTS

4.1 Implementation Overview

The revised abstract defines a supervised revenue-prediction and constrained-allocation system [4]. The conference draft identifies historical campaign inputs, preprocessing, a prediction stage, optimization, what-if analysis, dashboard presentation, and cloud deployment as the planned sequence [6]. The deployed user interface is the strongest evidence available for the present implementation: it shows a campaign configuration form, a budget optimizer page, analytics, scenario and report views, and additional demo pages [7].

The default configuration visible in the application is Omnichannel Growth Q3, Zoho, SaaS, India, Search Engine Marketing, 30 days, target revenue ₹75,000, and total budget ₹25,000. The India envelope shown is ₹10,000 to ₹40,000, split between Google Ads and Meta Ads with stated platform bounds. These are the application's pre-populated sample values, not observed results from a real campaign.

4.2 Prediction and Optimizer Implementation

The budget screen describes a Gradient Boosting ML model for baseline revenue and a platform-specific logarithmic response curve. It presents an SLSQP constrained optimizer with exact budget equality and min/max platform limits. The updated deployment completed the dashboard optimizer request. For the displayed ₹25,000 sample campaign, the model returned predicted revenue of ₹87,240 and a 3.49x revenue-to-budget ratio. The optimizer allocated ₹14,928 (60%) to Google Ads and ₹10,072 (40%) to Meta Ads; both platform bounds were respected. These are outputs from the displayed sample configuration, not measured results from a real campaign or evidence of predictive accuracy.

4.3 Interface Evidence

Figures 4.1-4.9 capture the deployed interface before the latest successful optimizer run. They document the page layout and default workflow; the live run output is summarized in Section 4.2. The A/B page describes a uniform baseline and an optimized variant but notes that the optimized allocation depends on live optimizer output. The audience page explicitly labels its profiles as synthetic demo data. The reports page shows a pending model run. The ad generator contains pre-filled SaaS copy and a preview; its presence alone does not establish that an external generative model was called.

4.4 Runtime Observations and Evaluation Plan

Check	Observation	Status
Dashboard load	Campaign overview rendered; current sample result displayed	Observed
Campaign setup	Inputs and India constraints rendered with sample defaults	Observed
Budget optimizer request	₹87,240 predicted revenue; 60/40 Google-Meta allocation; bounds respected	Completed
Analytics	Page rendered; optimizer output behavior not rechecked after the latest run	Observed
What-if workflow	No baseline campaign optimization; simulation requires live	Not completed

	request	
Audience and A/B pages	Demo labels visible; audience profiles marked synthetic	Demo interface only

4.5 Metrics Required for Final Evaluation

Measure	Definition / purpose	Use
MAE	Mean absolute difference between actual and predicted revenue	Interpretable average error
RMSE	Square root of mean squared prediction error	Penalizes larger errors
R-squared	Proportion of target variance explained on held-out data	Compare predictive fit
Constraint residual	Absolute difference between total budget and sum of allocations	Verify equality constraint
Scenario delta	Change in predicted revenue or allocation between budgets	Assess what-if sensitivity

These metrics are an evaluation plan, not reported outcomes. A credible final experiment should state dataset source and size, feature definitions, time-aware train/test split, baseline model, selected model, hyperparameters, and all measured results. Allocation quality should be compared with simple baselines such as equal split and historical allocation, while reporting feasibility and sensitivity to budget bounds.

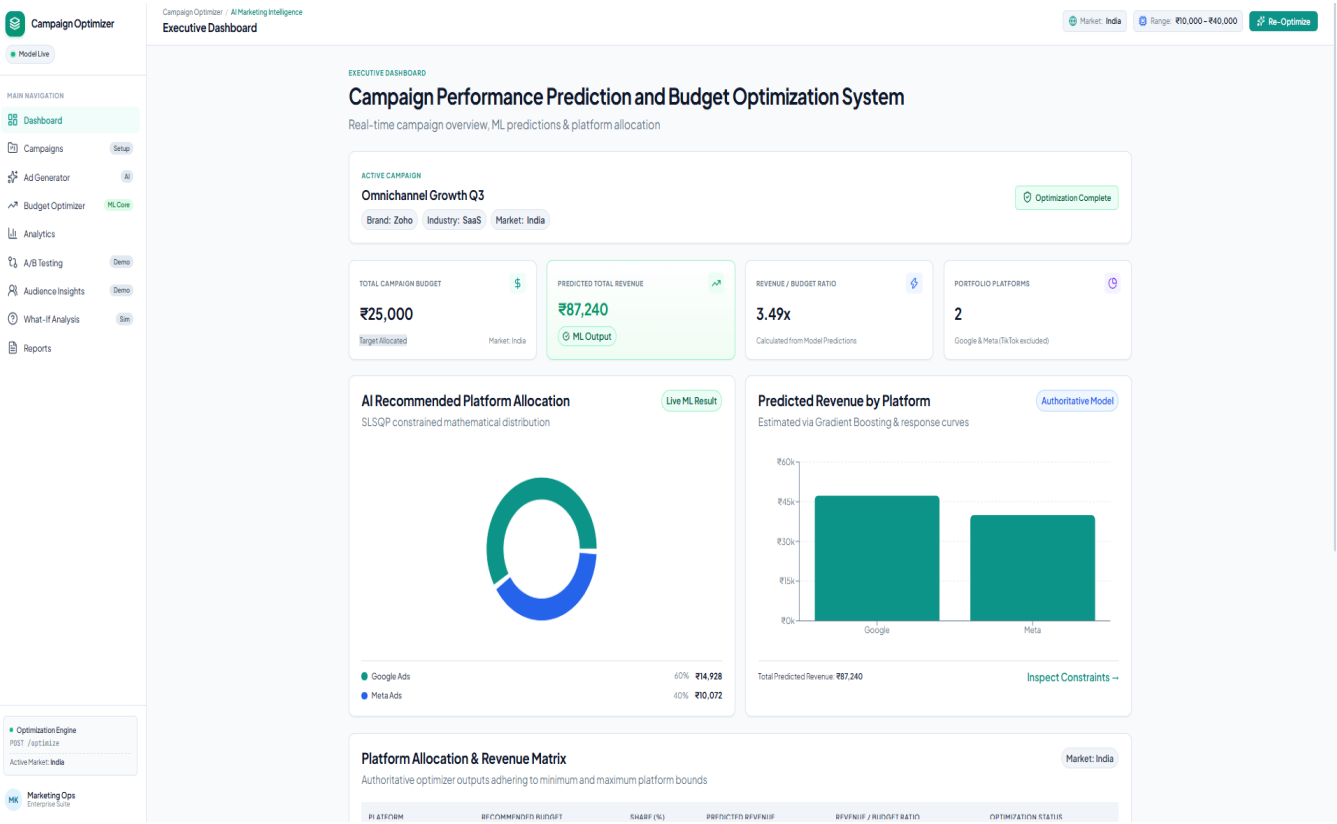


Figure 4.1: Executive dashboard showing the sample campaign, ₹25,000 configured budget, two-platform portfolio, and pending prediction fields.

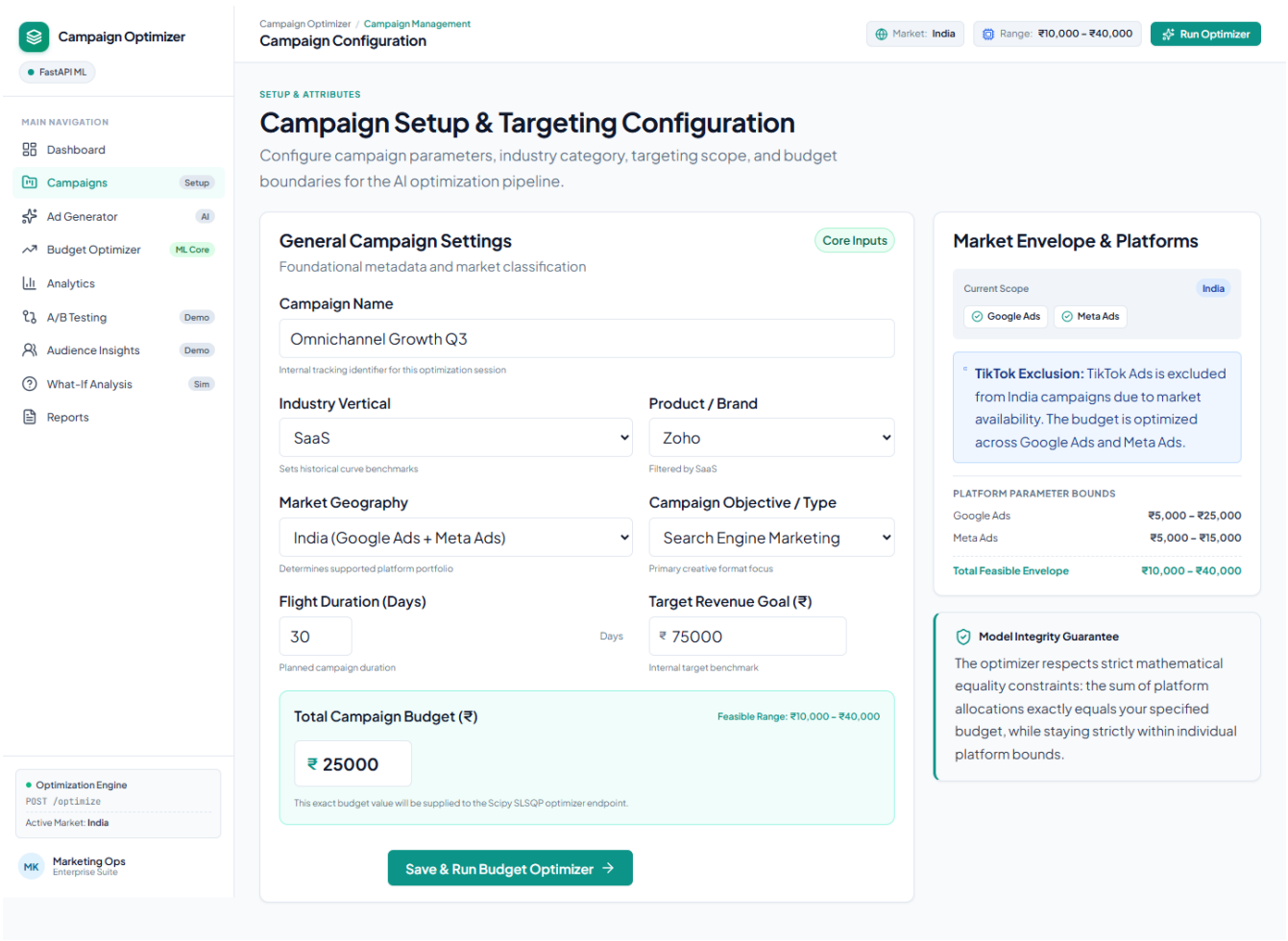


Figure 4.2: Campaign setup page showing campaign inputs, the India market envelope, platform bounds, and TikTok exclusion.

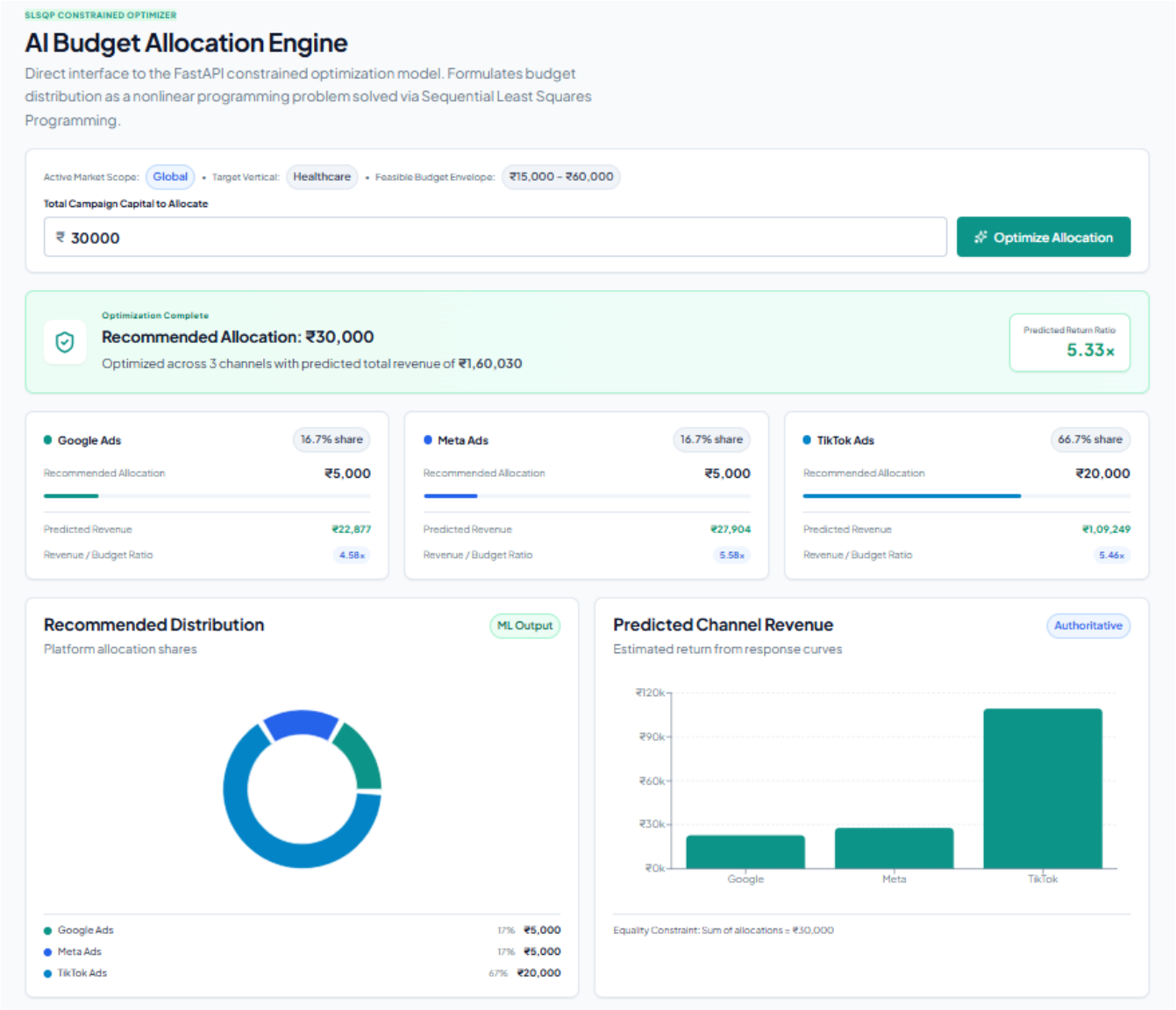


Figure 4.3: Budget optimizer page describing the SLSQP workflow, response curves, platform bounds, and ₹25,000 sample budget.

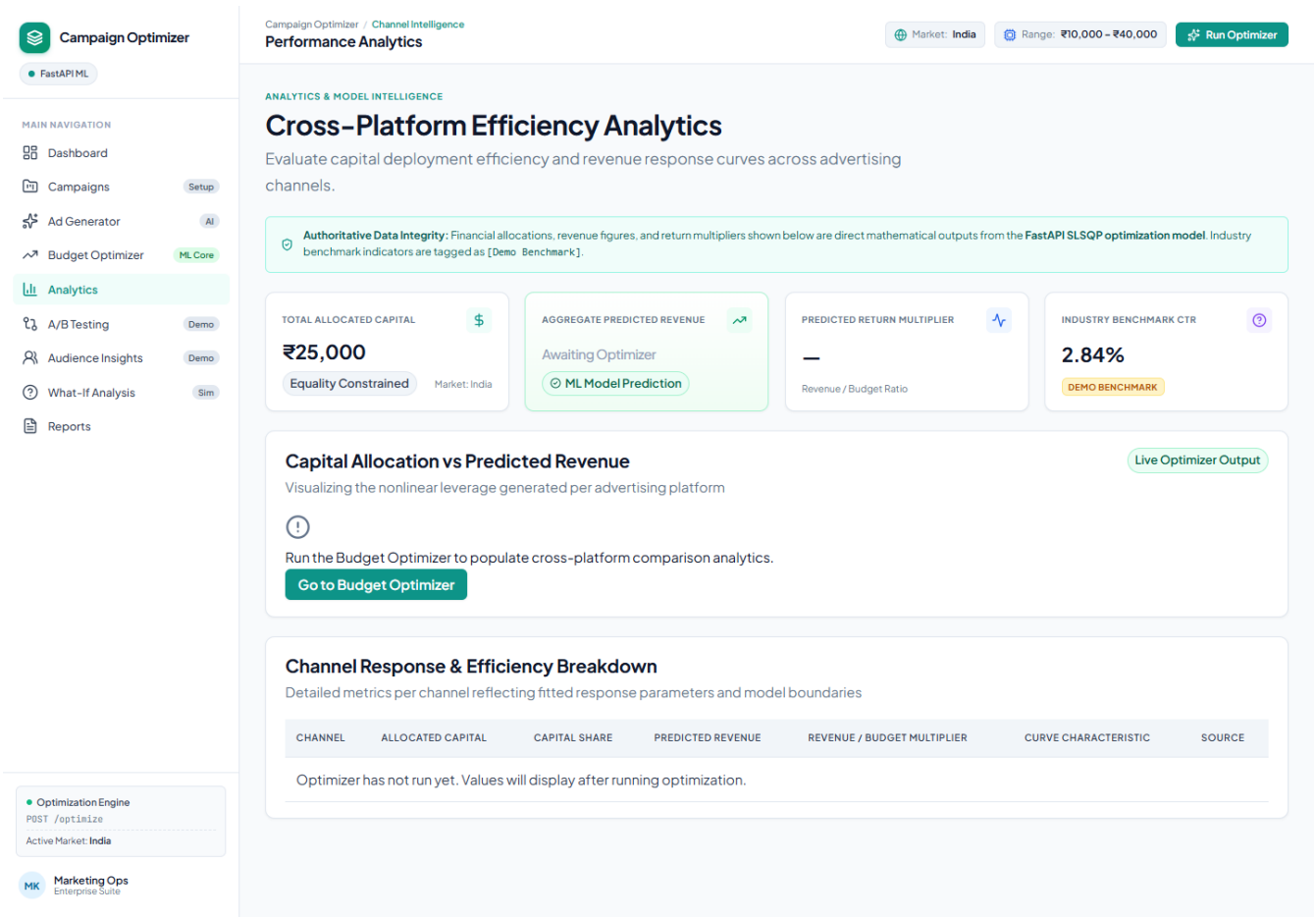


Figure 4.4: Analytics page with pending prediction output and an explicitly labeled demo benchmark CTR.

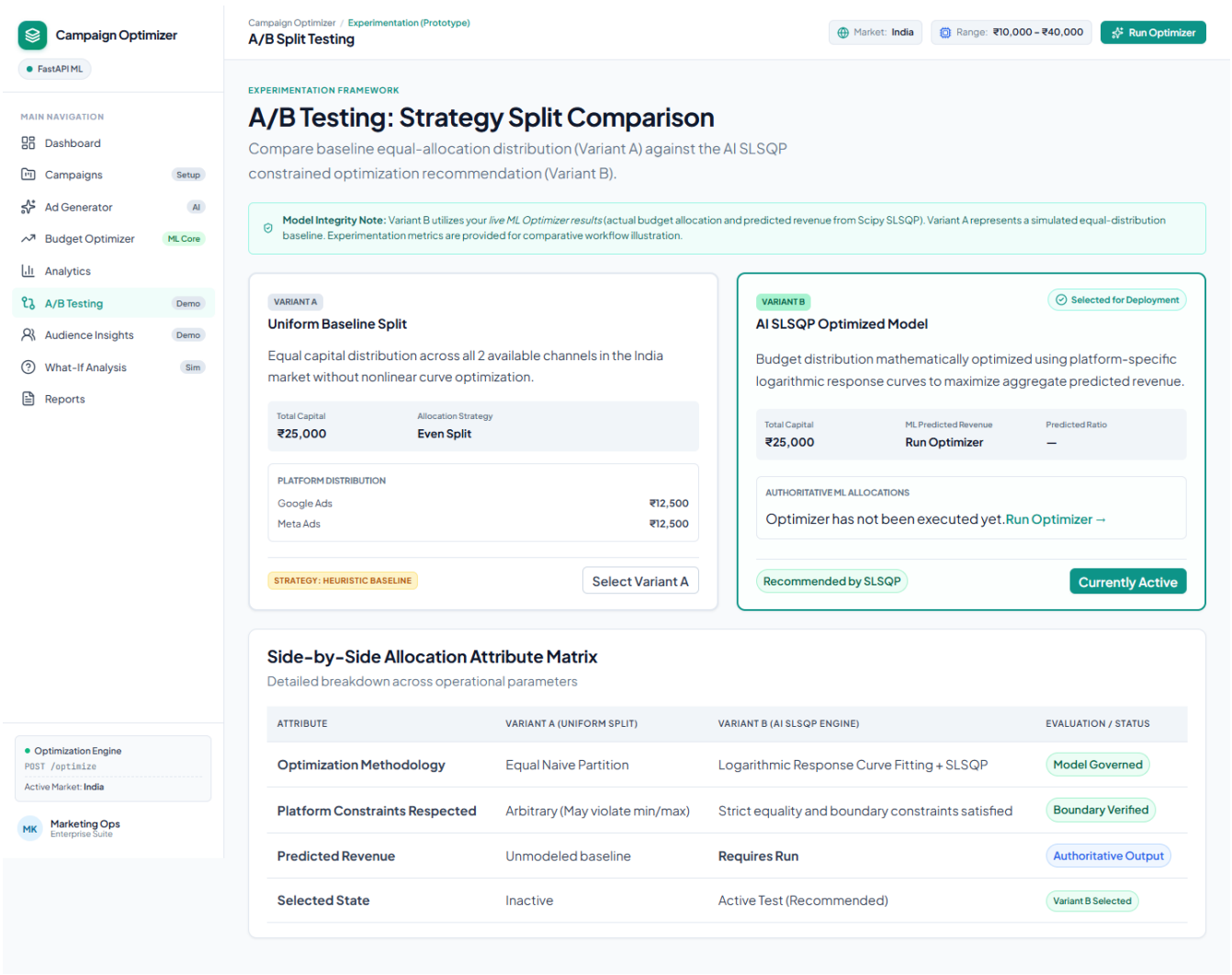


Figure 4.5: A/B split comparison page. The optimized variant depends on live optimizer results; its metrics are presented as workflow illustration.

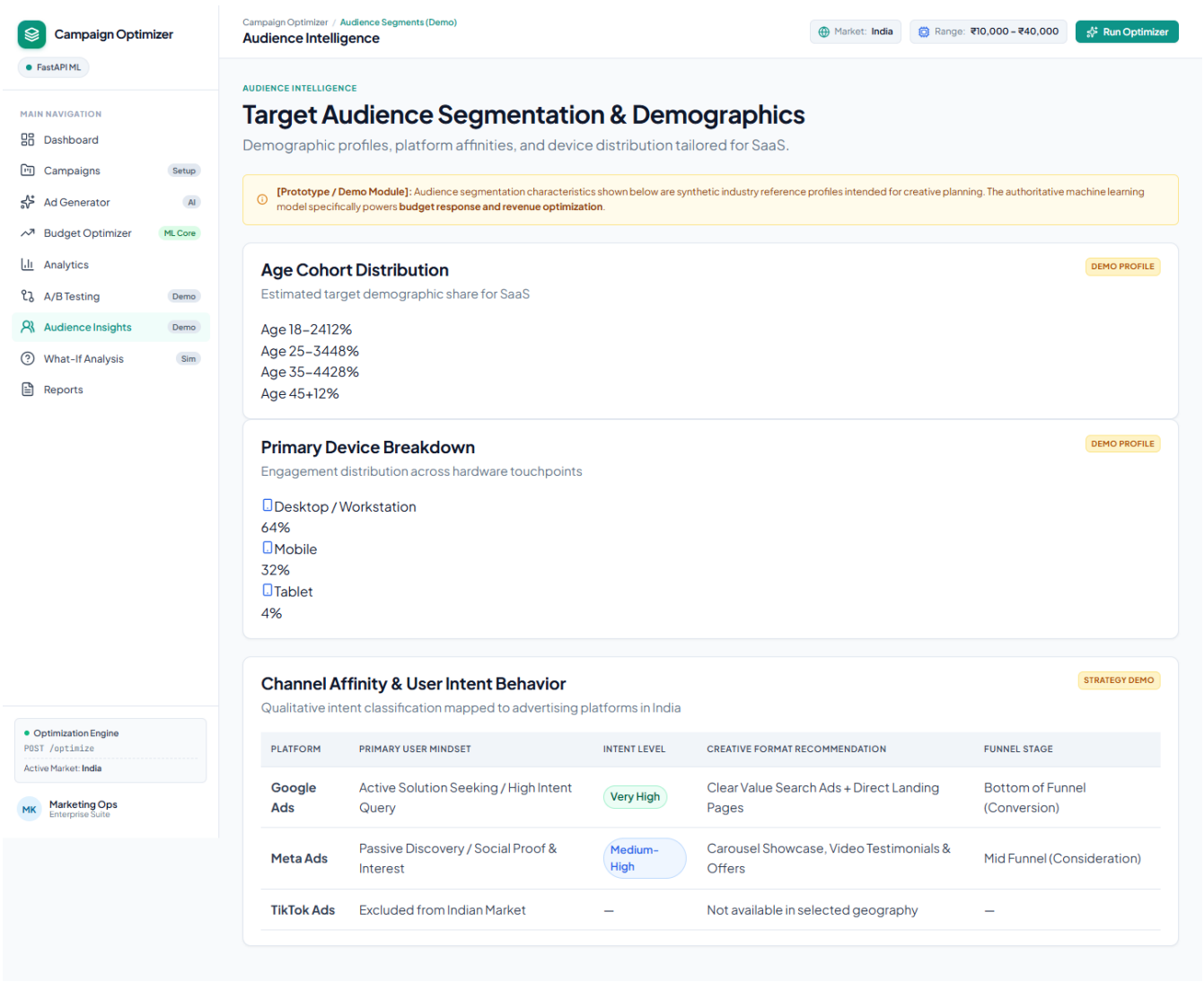


Figure 4.6: Audience segmentation page. The visible notice identifies its demographic profiles as synthetic industry reference data.

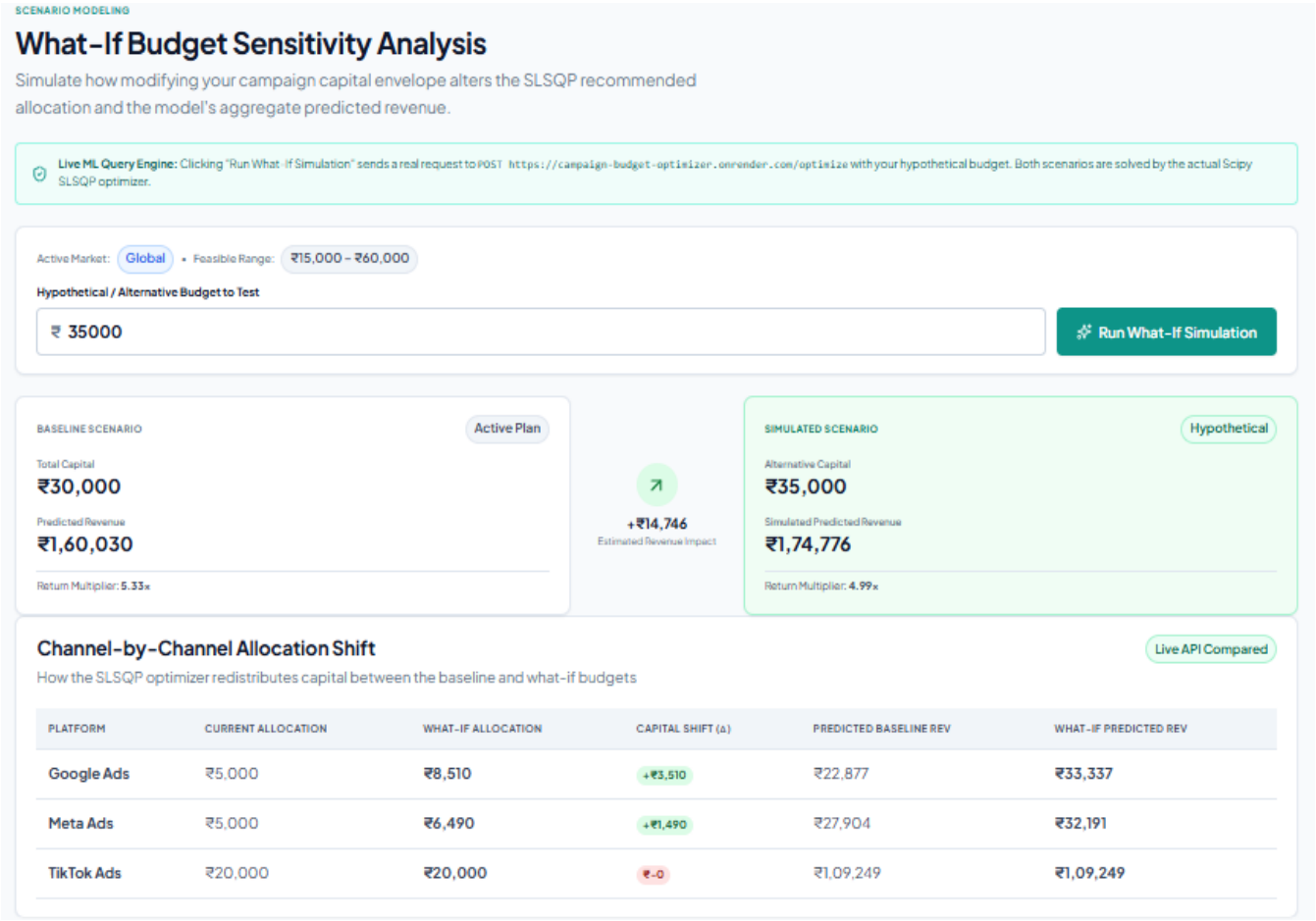


Figure 4.7: What-if budget simulator showing the ₹35,000 sample scenario and the requirement to run a baseline first.

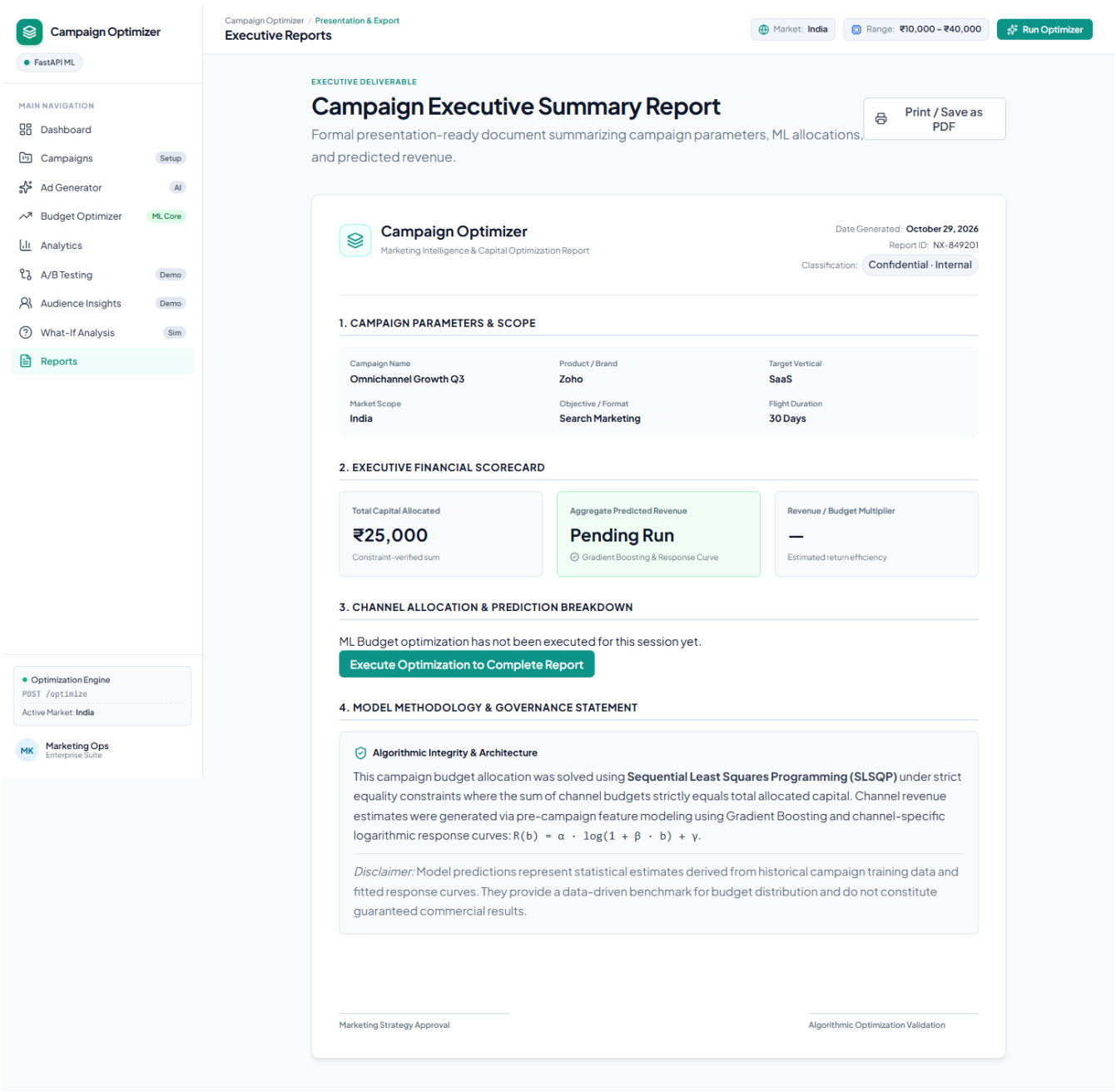


Figure 4.8: Executive report view with a pending revenue prediction and an action to execute optimization.

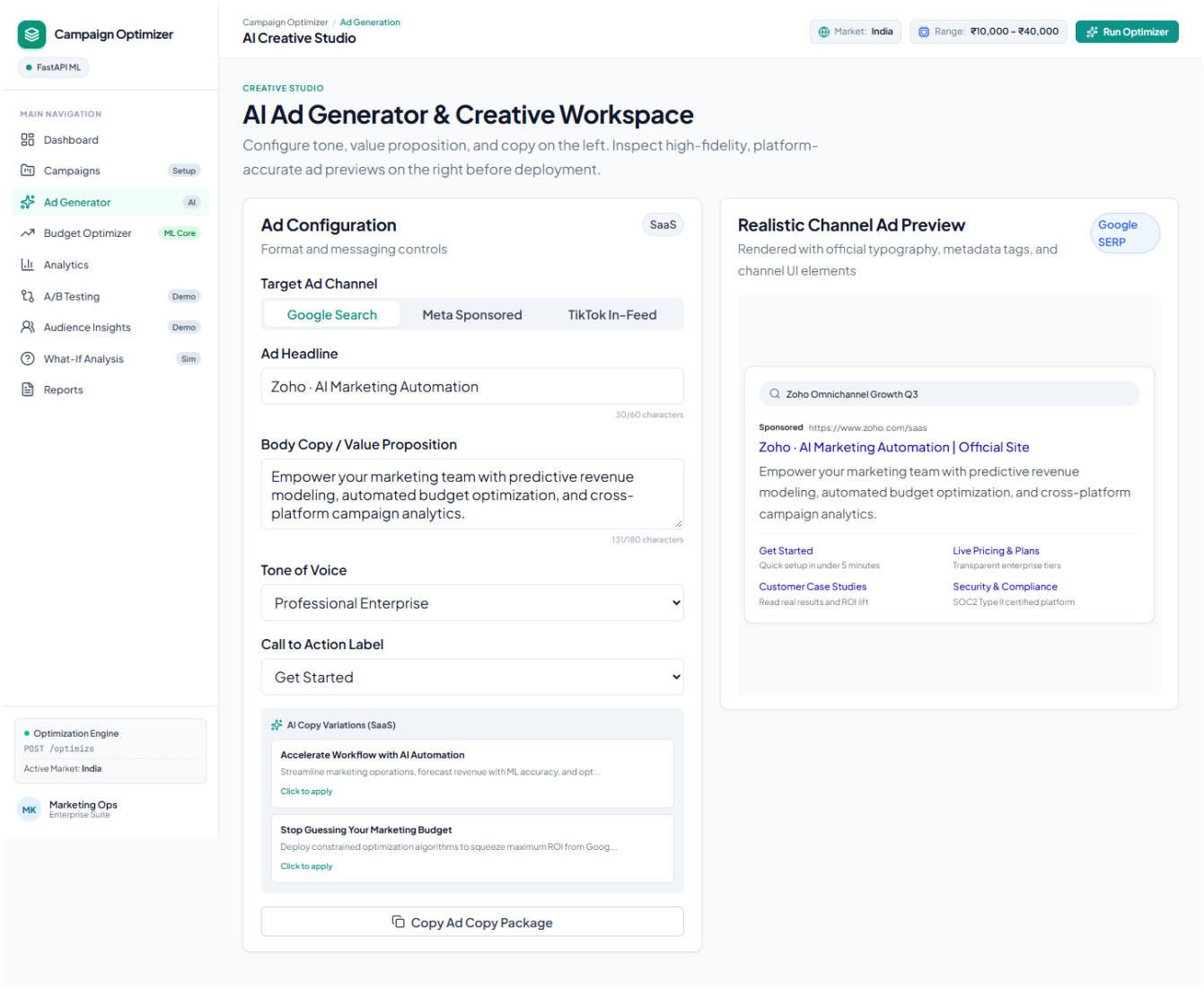


Figure 4.9: Ad creative workspace with pre-filled SaaS copy, channel controls, and a Google Search preview.

CHAPTER 5 - CONCLUSION AND FUTURE WORK

5.1 Conclusion

The project brings campaign revenue prediction and constrained budget allocation together as a marketing decision-support workflow. Its final abstract names the Gradient Boosting Regressor and what-if analysis. The deployed interface shows campaign setup, India-specific platform constraints, an optimizer page describing SLSQP, analytics, demo comparisons, and report presentation. The supplied base paper provides research context for using predicted campaign outcomes to support allocation decisions [1, 4-7].

The updated deployment returned a live sample prediction and budget allocation during inspection. The report makes no claim about model accuracy, revenue uplift, or the optimizer's real-world benefit; those require documented training data and measured evaluation. The interface defaults document what the application is configured to present; measured performance must come from a completed backend run and a reproducible dataset evaluation.

5.2 Project Contribution

The practical contribution is the integration of a campaign form, predictive and optimization workflow, market constraints, scenario interface, and report-oriented dashboard in one accessible prototype. This contribution is appropriately framed as a student implementation of established research ideas, rather than a new advertising-allocation algorithm.

5.3 Future Work

- Record the deployed FastAPI endpoint version and the exact request and response schema for reproducibility.
- Document the historical dataset, record count, date coverage, target variable, feature timing, and data-cleaning decisions.
- Train and evaluate the Gradient Boosting model with a time-aware held-out set; report MAE, RMSE, and R-squared, with a simple baseline.
- Validate response-curve fitting and units; inspect whether channel revenue curves are monotonic and whether parameter values are calibrated on held-out campaigns.

- Test optimizer feasibility at lower, upper, and boundary budgets, including budgets where platform bounds make an allocation infeasible.
- Run controlled what-if scenarios and report allocation and predicted-revenue changes without presenting them as causal effects.
- Separate live model outputs from demo benchmarks and synthetic audience profiles in the UI and in exported reports.
- Add automated tests for input validation, constraints, error handling, API timeouts, and report generation.
- Document data retention, campaign confidentiality, model versioning, and cloud deployment configuration.

5.4 Final Summary

The application presents a clear path from campaign context to allocation review. Its next evidence milestone is not a larger claim; it is a repeatable run with documented data and verifiable outputs. Once those are available, this report can be updated with measured model results, allocation tables, and completed what-if comparisons.

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APPENDIX A - REQUIREMENTS AND OBSERVATION NOTES

A.1 Sample campaign configuration visible in the deployed UI

Field	Displayed value	Interpretation
Campaign	Omnichannel Growth Q3	Interface default
Brand / industry	Zoho / SaaS	Interface default
Market / objective	India / Search Engine Marketing	Interface default
Duration / target	30 days / ₹75,000	Target is not predicted revenue
Total budget	₹25,000	Within displayed feasible envelope
Platforms	Google Ads and Meta Ads	TikTok excluded in India
Alternative scenario	₹35,000	What-if input shown; simulation not completed

A.2 Interpretation of interface evidence

A rendered page verifies that a screen and its displayed labels were available. It does not by itself verify server-side execution, model quality, data provenance, causal lift, or production readiness. Screens marked Demo or Prototype in the interface are treated accordingly. The updated deployment returned a sample prediction of ₹87,240 and a Google/Meta allocation of ₹14,928/₹10,072; these displayed outputs do not establish model quality or real-world lift.

A.3 Reproducibility and acceptance checks

A complete evaluation should record the dataset version, prediction target, feature timestamps, train/test split, model version, prediction metrics, optimizer constraints, request and response payloads, and budget scenario. The same input should produce a documented result under the same model and configuration. Demo benchmarks and synthetic audience profiles should remain clearly separated from live campaign outputs.